

WASP-177b

R-2423

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-177_200714 · created 2026-08-21T23:52:38+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459044.8949 +/- 0.0056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.049
Transit depth (Rp/Rs)^2	2.87 +/- 1.66 [%]
Orbital Inclination (inc)	88.91 +/- 2.89
Airmass coefficient 1 (a1)	1.74 +/- 1.17
Airmass coefficient 2 (a2)	-0.6 +/- 1.67
Scatter in the residuals of the lightcurve fit is	63.64 %
Best Comparison Star	#1 - [323, 406]
Optimal Aperture	1.58
Optimal Annulus	6.0
Transit Duration (day)	0.09 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.169 ± 0.049 vs 0.1618 (deviation 0.15σ)
Duration fit vs published	0.09 d vs 0.0721 d (deviation 1.12σ)
Mid-time O–C	-7.9 min
Depth SNR	3.4
Residual scatter	63.64 %

Light curve

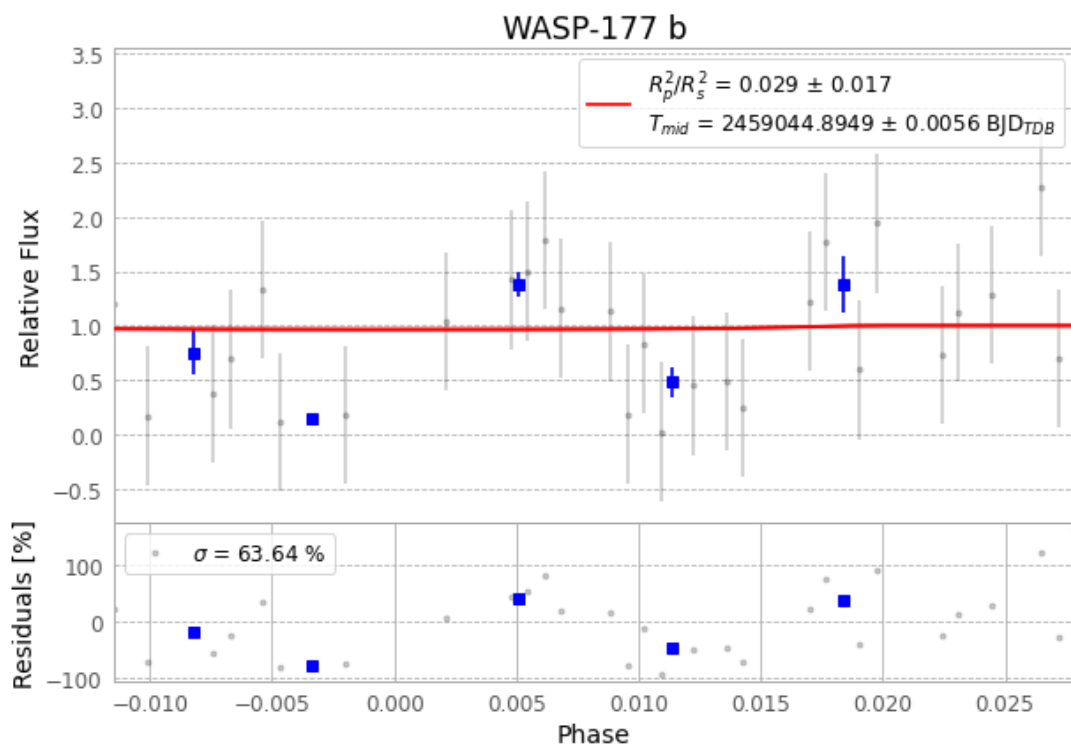


Figure 1 — FinalLightCurve_WASP-177 b_2020-07-14.png (SHA-256 ddf669a64a035a7d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [431, 476], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d4f77412cf0050f1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

60 FITS frames (39 MB), WASP-177200714082712.FITS → WASP-177200714112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-177-200714.log (dd0f80868a1b...), FinalLightCurve_WASP-177 b_2020-07-14.png (ddf669a64a03...), FinalLightCurve_WASP-177 b_2020-07-14.csv (fc6968e66140...)

Compute resources

Wall time 122.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 23:52Z.